Self – Questions:

1. Design and implement an iterative TCP server for an Online Railway Reservation System that serves one client at a time. The server should maintain a database of trains, routes, available seats, fares, and booking details. A client should be able to search trains, check seat availability, book tickets, cancel reservations, and view booking status. Since the server is iterative, it must complete all operations for one client before accepting the next client connection. The server should validate all requests, update the reservation database after every transaction, maintain a log of client activities with timestamps, and handle invalid booking requests gracefully.
2. Develop an iterative TCP server for a Hospital Appointment Management System. The server should maintain doctor schedules, available appointment slots, patient records, and consultation timings. Clients should be able to register, book appointments, cancel appointments, search doctor availability, and view appointment history. The server should process one client's complete request before serving the next client, ensure that appointment slots are updated correctly after each transaction, generate confirmation messages, and record all operations in a log file while handling invalid inputs appropriately.
3. Implement an iterative TCP server for a Movie Ticket Booking System. The server should maintain information about movies, show timings, available seats, and ticket prices. A client should be able to view the movie list, check seat availability, book tickets, cancel bookings, and display booking details. The server should process one client request at a time, update seat availability after every successful transaction, and provide appropriate messages for successful or unsuccessful operations.
4. Develop an iterative TCP server for a College Library Management System that processes one client at a time. The server should maintain a catalog of books containing Book ID, Title, Author, Category, and Availability Status. Clients should be able to search for books by title or author, check availability, issue books, return books, and view issued book details. The server should validate all requests, update the library records after each transaction, and terminate the client connection gracefully before accepting the next client.

Exercise Questions to be implemented:

Easy (E)	Medium(M)	

Self – Questions:

1. Design and implement a concurrent TCP server for an Online Examination Management System that can simultaneously handle multiple student clients using multithreading or multiprocessing. Each client must authenticate using a unique username and password before accessing the system. After successful authentication, the client should be able to retrieve examination questions, submit answers, and view the final score after evaluation. The server should maintain separate sessions for each client, prevent data corruption when multiple clients access the question database simultaneously, log all client activities with timestamps, and gracefully handle client disconnections and invalid requests.
2. Develop a concurrent TCP server for a Hotel Reservation System that supports multiple clients booking rooms concurrently. Each client should be able to view room availability, book a room, cancel a reservation, and check booking details. The server must ensure that two clients cannot reserve the same room simultaneously by implementing appropriate synchronization mechanisms (such as mutexes, semaphores, or synchronized blocks). All booking transactions should be stored permanently in a file or database, and the server should maintain detailed logs of every client request while handling concurrent connections efficiently.
3. Implement a concurrent TCP server that acts as a File Sharing Server capable of serving multiple clients simultaneously. Each client should be able to request the list of available files on the server and download a selected file. The server should create a separate thread or process for each connected client, allowing multiple file transfers to occur concurrently while ensuring reliable data transmission and proper connection termination.
4. Develop a concurrent TCP server for a Chat Notification System where multiple clients can connect simultaneously and exchange text messages through the server. The server should receive messages from one client and broadcast them to all other connected clients while maintaining individual client sessions. The application should detect client disconnections, remove inactive clients from the active client list, and continue serving the remaining clients without interruption.

Exercise Questions to be implemented:

Self – Questions:

1. *Implement the Go-Back-N Sliding Window Protocol to simulate reliable data transmission between a sender and a receiver. The program should allow the user to specify the window size, number of frames, and the frame to be lost or corrupted during transmission. Simulate the retransmission of frames upon timeout and display the sequence of transmitted, acknowledged, and retransmitted frames. Finally, analyze the total number of transmissions required for successful data delivery.*
2. *Develop a program to simulate the Selective Repeat Sliding Window Protocol for reliable data communication. The application should accept the window size, number of frames, and simulate random frame loss during transmission. The receiver should acknowledge only the correctly received frames, while the sender retransmits only the lost or corrupted frames instead of the entire window. Display the status of each frame, acknowledgment sequence, retransmissions performed, and compare the efficiency of the protocol in terms of total transmissions and successful frame delivery.*

Exercise Questions to be implemented: